Group Assignment

CT070-3-M-NDP

Network Design Performance

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1.0 Introduction

St. John's University (SJU) to an established higher education institution, founded in 1994, serving more than 15,000 students on a dispersed 150-acre campus featuring six principal structures plus one for research labs with significant academic and administrative staff [50].

The Cyber Security & Digital Forensics Lab (CSDFL) is an important laboratory at SJU designed for cutting-edge cybersecurity training, forensic investigations and applied research. The lab supports a broad range of users: Approximately 440 individuals can be found in an average week, including over 350 undergraduate students, more than 50 postgraduate students, over 10 PhD researchers, around 20 academic staff and around 10 administrative clerks. Its work involves handling large forensic disk images that frequently exceed 100GB, processing packet captures, and managing virtualised environments — all workflows that require high-speed, reliable network connectivity.

The ICT team performed a recent application performance assessment that highlighted several shortcomings in the current LAN infrastructure, including low throughput/throughput constant/microbursts/high latency/packet loss/jitter/unreliable connectivity/bandwidth congestion. These vulnerabilities have affected forensic imaging workflows, as well as slowing down research simulations and decreasing academic and research productivity.

So, in response — after consulting with university senate, academic staff and students the ICT team has undertaken a full LAN redesign project. Part 1 of the Network Development Life Cycle (NDLC), this document addresses four foundational pillars of the design planning phase for the proposed upgrade:

- Top-Down vs Bottom-Up: Exploring Different Network Design Approaches
- Business and Technical Objectives — defining goals & targets
- Network Model — The analysis (Comparison and Selection of Best Architecture)
- Network Design Tools — Inter- and intra-network area network design

Collectively, these sections lay the analytics groundwork for a durable, scalable, and high-speed LAN solution to meet CSDFL current needs as well as future demands.

2.0 Network Design Approaches

The initial step in the NDLC is to choose a plan organisation that best fits your exact problem. In over-all, there are two styles: Top-Down and Bottom-Up. These choices take diverse features, benefits and limits that must be measured against the CSDFL's exact requests.

2.1 Top-Down Network Design

Before any technology is chosen, the Top-Down approach starts with understanding and realizing the organization's goals, requirements, and limits. The rationale behind this model is that the network should serve the lab rather than the other way around where available equipment shapes how you build a network. For the CSDFL, this means first specifying security requirements, user group needs, performance expectations and scalability goals, and then selecting technologies that meet those parameters. This approach is one described by ***Oppenheimer (2011)*** wherein understanding the requirements of the users and applications on the network, before going into physical design.

2.1.1 Advantages

- Goal Alignment: The lab's specific goals of a secure, high-speed, and future-proof network are considered in every decision.
- Security by Design: (The lab deals with sensitive digital evidence and forensic data, so controls like encryption, firewalls, and access control are built in rather than making the system secure after deployment ***(Casey, 2011; Joint Task Force, 2020).***
- Scalability planning — ***Oppenheimer (2011); Stallings (2016)*** reported that long-term growth such as upscaling students' enrolment, devices or research workloads is explicitly planned in the design phase.
- Long-Term Viability: Planning ahead for intended use makes sure the network continues to work well as institutional usage adapts ***(Kurose & Ross, 2022).***

2.1.2 Disadvantages

- time investment: Groundwork in the form of a full needs analysis and aims specification can prolong the initial stage of planning ***(Oppenheimer, 2011)*** before hardware selection.

- Risk of Leaving Out Some Technical Detail: Being too high-minded or focused on goals may lead to underplaying some constraints, but this risk is small if developers cross-consult rigorously ***(Forouzan, 2013).***

2.2 Bottom-Up Network Design

This Bottom-Up approach scale starts with what's technically available on hand by selecting routers, switches and hardware devices to supply the physical infrastructure alongside the creation of a network that meets organisational needs. It is technology-driven instead of being requirements-driven.

2.2.1 Advantages

- Rapid Initial Deployment: Deploying the method can begin immediately as it uses existing/homemade hardware and does not require longer planning stages ***(Stallings, 2016).***
- Technical Grounding: The architecture is established through proven technical standards, minimizing uncertainties regarding integrations during deployment ***(Forouzan, 2013; Kurose & Ross, 2022).***
- Hardware Familiarity: The engineers that are deploying the selected hardware are already familiar with it, allowing them to deploy it in a time efficient manner, removing onboarding time ***(Stallings, 2016).***

2.2.2 Disadvantages

- Looking out: Out of alignment with organisational needs — without goal-setting at the outset, networks may fail to meet operational needs of the lab ***(Oppenheimer, 2011).***
- Lack of Scalability: Networks that are built around the current hardware capabilities may prove to be counterproductive, as a costly redesign may be necessary if the network needs to scale in the future ***(Kurose & Ross, 2022; Stallings, 2016).***
- Security as an Afterthought: Security is an afterthought and often added post-deployment, which is problematic particularly in an environment that deals with sensitive forensic evidence and where a breach could affect chain of custody ***(Casey, 2011; Joint Task Force, 2020).***

2.3 Comparative Analysis

Dimension	Top-Down	Bottom-Up
Starting Point	Organisational goals and requirements	Available hardware and technology
Security Integration	Built in from the start	Added after hardware selection
Scalability Planning	Explicitly addressed upfront	Limited; requires redesign to expand
Deployment Speed	Slower initial phase	Faster initial deployment
Best For	Complex, security-critical environments	Small, low-complexity deployments
Suitability for CSDFL	**High — aligns with security and research needs**	**Low — insufficient for 440+ user, security-critical environment**

Table 1 - Top-Down and Bottom-Up Design Approaches Comparison

2.4 Recommended Approach for CSDFL

The CSDFL LAN redesign follows TOP-DOWN design methodology. The lab works within a security-sensitive environment, uses sensitive forensic data, and supports a heterogeneous user population with different access requirements. Such characteristics call for a top-down, needs-first design process that has security, performance and scalability built in from the start. Network design must be in sync with organisational goals to result in viable long-term value ***(Oppenheimer, 2011).*** Though sufficient for the small or low-complexity deployments, The Bottom-Up approach lacks the strategic depth necessary for deployment in a high-density and security-sensitive research environment such as the CSDFL.

3.0 Business and Technical Objectives

Following the Top-Down design method, this section identifies the business and technical goals that must be met by CSDFL LAN redesign. These objectives provide the measurable foundation upon which the proposed solution will be judged.

3.1 Organizational Context

Large forensic disk images, packet captures and virtualised environments that require high-speed data transfer and stable connectivity are handled by the CSDFL. Before this project began, an ICT assessment was already conducted by ICT and MSSW to identify some of the major LAN limitations including low throughput, high latency, packet loss, unreliable connectivity and bandwidth congestion. Such vulnerabilities affect forensic imaging, prolong simulations, and increase research unproductivity. The existing infrastructure is inadequate given the steep increase in student enrollments and research activities. A high-speed LAN redesign to make it more performance-driven, secure, reliable and scalable.

3.2 Business Objectives

3.2.1 Improve Academic and Research Productivity

High performance networking should be vital when dealing with forensic images > 100GB Network throughput and latency have directly impact to application performance ***(Kurose & Ross, 2022, Chapter 1).*** Through a 10Gbps backbone, transfer times will vastly improve and virtual labs will be much more responsive. As stated by ***Oppenheimer (2011)*** for network design to be effective they need to align with organisational goals and as such support research continuity directly.

3.2.2 Enhance Institutional Reputation

It makes our institutions more competitive, because it ensures their credibility. Thus ***ISO/IEC 27001 (2022a)*** assigns the structured information security governance as basis for trust and funding opportunities from stakeholders.

3.2.3 Rise Operational Efficiency

Hierarchical network styles decrease working difficulty and make simpler troubleshooting ***(Cisco Systems, Inc., 2020).*** Upgrading decreases lost time and recovers Mean Period to Patch-up (MTTR), serving keep teaching and explore on track.

3.2.4 Data Safety and Compliance, Reinforced

The laboratory controls sensitive electronic signatures.Consistent with ***NIST SP 800-53 (2020)***, control objects are made in layers and tight checks carried out for access control and System & Communications Protection.According to ***IEEE 802.1X (2020)***: It proposes port-based network access control that could the LAN-port side of an Ethernet switch to be shut down in such a way as if all ports on this new switch were dead thereby accurately controlling which is connected or disconnected for stations you provide connectivity to from your locations.

3.2.5 Confirm Scalability and Future Development

Module architecture allows for growth without redesign (***Stallings, 2016***). Existing LAN will no longer be able to handle growing enrolment, more devices or expanding research workloads.

3.3 Technical Objectives

3.3.1 High-Speed Backbone Infrastructure

A dedicated 10Gbps fibre backbone connecting all core, distribution and access layer switches which complies with the IEEE 802.3 family of standards ***(IEEE, 2018)*** is suggested in the CSDFL redesign. This specification is directly guided by the lab's forensic workload: transferring disk images greater than 100GB over 1Gbps adds transfer times of greater than 13 minutes, creating unacceptable delays for time-sensitive investigations. An example backbone, at 10Gbps, brings this down to less than 90 seconds directly serving research continuity ***(Kurose & Ross, 2022).*** High bandwidth also minimizes congestion and packet loss, thus improving Quality of Service for simultaneous virtualised environments over 440+ active clients ***(Cisco Systems, Inc., 2020; Forouzan, 2013).***

3.3.2 Hybrid Hierarchical Network Architecture

The CSDFL's micro-data centre will employ a Hybrid Hierarchical architecture, where a three-layer Core–Distribution–Access backbone is integrated with an SDN-controlled Spine-Leaf fabric ***(Cisco Systems, Inc., 2020)*** that enables structured fault isolation of the sub-networks as data flows and centralized policy enforcement for East-West high-throughput traffic flows ***(Oppenheimer, 2011).*** Section 4 discusses the comparative analysis and architectural rationale for this choice.

3.3.3 Network Segmentation and Micro segmentation Controls

Network segmentation segments a network into separate logical areas to control the flow of traffic, enforce least-privilege access and limit security incidents ***(Forouzan, 2013; Kurose & Ross, 2022)***. A conventionally utilized approach conducted by VLANs at Layer 2 while subnet routing exists on Layer 3, this is swiftly becoming seen as inadequate to operate as an isolated method. VLANs do not provide some guarantee for security at the endpoint: if a device gets access to its segment — through port misconfiguration, credential misuse, or Rogue Device — it automatically receives an implicit trust relationship and can move laterally without further challenge. This is an infeasible risk model in a Digital Forensics laboratory, where the integrity of evidence can be subject to legal pursuits ***(Casey, 2011; Joint Task Force, 2020; Rose et al., 2020).***

Thus, the CSDFL redesign leverages a microsegmentation based on a Zero Trust Architecture (ZTA) crictique as per NIST SP 800-207 ***(Rose et al., 2020).*** With ZTA, where a device is no longer trusted based on internal network location — authentication and authorisation are independent per-session functions for each identity-verified policy enclave to check before allowing any pass-through traffic between zones. Enforcement of the five logical zones (i.e., Student, Staff, Forensic Research, Evidence Server and Management) through SDN-controlled policy rules applied at the Distribution Layer rather than static ACLs alone are used to enforce the logical zones with evidence server zone having a strictly default-deny posture ***(ISO/IEC, 2022b; Rose et al., 2020).***

This architecture fulfills both ISO/IEC 27002:2022 Control 8.22 ***(ISO/IEC, 2022b),*** which requires risk-based segregation of networks with documented and auditable explanation for traffic, as well as CIS Controls v8. 1/ 12.2 and 12.8 - Segmentation (least privilege) / Physically or logically isolated administrative plane ***(Center for Internet Security, 2024).*** So the Management VLAN is air-gapped from the primary network and has no internet connectivity. IEEE 802.1X ***(IEEE, 2020)*** offers access-layer identity binding, preventing un-enrolled devices from claiming VLAN membership — in combination achieving a defence-in-depth model acceptable for an environment where the impact of segmentation failure has both operations & legal ramifications.

3.3.4 Redundancy and High Availability

These are redundancy devices, such as stacked core switches, redundant fibre uplinks, LACP and Uninterruptible Power Supplies (UPS). Joint, these devices are designed to reach 99.9% availability.

3.3.5 Quality of Service and Network Monitoring

Together with central logging, continuous monitoring using SNMP allows proactive fault detection ***(Joint Task Force, 2020).*** Under heavy loads, QoS policies may also give priority to forensic and investigative traffic in order to keep performance stable***(Forouzan, 2013).*** fault detection is made possible via continuous monitoring through SNMP and centralised logging ***(Joint Task Force, 2020).*** QoS policies can also prioritise forensic and research traffic to keep performance stable under load ***(Forouzan, 2013).***

3.4 Key Performance Indicators (KPIs)

KPI	Target	Standard	Reference
Network Throughput	1–10 Gbps	IEEE 802.3	IEEE, 2018
Internal Latency	< 5 ms	Enterprise Network Design	Oppenheimer, 2011; Kurose & Ross, 2022
Network Availability	≥ 99.9%	Enterprise Network Design	Oppenheimer, 2011; Kurose & Ross, 2022
Packet Loss	< 1%	IEEE 802.3	Forouzan, 2013
Concurrent User Support	200+ users	CSDFL Requirement	ICT Assessment
Forensic Image Transfer	100GB+	CSDFL Requirement	ICT Assessment

Table 2 - Key Performance Indicators

4.0 Network Model Analysis

Now that we've established our business and technical objectives, this section assesses four candidate network architectures to determine which model is most suitable for the CSDFL. Before a final recommendation is made, each model is evaluated against the lab's needs.

4.1 Three-Tier Hierarchical Model

A hierarchical strategy is a multilevel system that splits the network into three layers: Core, Distribution, and Access. This classical offers a deterministic parting of complexity in such a high density environment as the CSDFL and avoids local faults becoming global outages (***Oppenheimer, 2011).***

The Access Layer is the first line for legal workstations and analysis kiosks where it controls initial traffic ingress and basic security tagging. The Distribution Layer is the policy enforcement pivot, aggregating traffic and processing inter-VLAN routing as well as applying ACLs and Qos rules prioritising forensic data. The Core Layer, a high-speed backbone - ideal for fast packet switching without the latency implicated in complex policy checks - ensures everything from forensics images larger than 100GB can be wire speed transferred to storage arrays. Such modularity allows the lab to accommodate more switch blocks without having to make changes to the core system as more users onboard beyond 440 ***(Cisco Systems, Inc., 2020; Kurose & Ross, 2022).***

4.2 Flat Network Architecture — Limitations

In a flat network configuration, all the devices are connected in a shared communication domain. This model is easy to configure and also cheap if you have a small deployment but is inherently broken in dense environments. However, as endpoints scale, broadcast traffic becomes multiplicative and every node has to process each broadcast — this causes performance degradation until critical forensic tasks cannot be completed ***(Hirsi et al., 2024).***

Security is no different: without segmentation, one compromised device can spread malware at will to every other connected system. Flat architectures ***(Hirsi et al., 2024)*** are not sustainable in a secure research environment and comply with regulation simply because by definition it lacks regulation.

4.3 Spine-Leaf Architecture

The Spine-Leaf architecture leverages a two-tiered design — spine routers that create a central mesh, and leaf switches along the edge, where every leaf connects directly to each spine offering: In this configuration any two endpoints will communicate in exactly two hops. As a result, this architecture is capable of supporting East-West server-to-server data flows much more efficiently than hierarchical models originally created to pass North-South user-to-server traffic ***(Al-Fares et al. 2008).***

Spine-Leaf provides lower and more predictable latency than hierarchical designs since it guarantees a fixed two-hop path between any two endpoints, while Equal-Cost Multi-Path (ECMP) routing spreads load evenly across all non-blocking spine links to ensure that none ever become bottlenecks ***(Al-Fares et al., 2008).*** For the CSDFL's micro-data centre — where more than 100GB of forensic images need to be passed between storage nodes — this architecture allows for a considerable performance improvement.

4.4 Software-Defined Networking (SDN)

SDN moves the network's control plane apart from its data plane, facilitating centralised and programmable management through a software controller. Instead of manually configuring every port separately, administrators can implement security regulations in a dynamic global way ***(Hirsi et al., 2024).***

For the CSDFL, SDN allows for an automated forensic artefact collection across a network with chain of custody. In the event of malicious activity being detected, the SDN controller has the ability to instantly quarantine compromised workstations in real time without manual intervention ***(Hirsi et al., 2024; Jiménez et al., 2024).***

4.5 Hybrid Model and Migration Strategy

Hybrid model combines several architectural models — would be something like a standard hierarchical architecture with SDN-driven Spine-Leaf fabrics that provide high flexibility and gradual phasing out of legacy systems. HARMLESS (which emulates SDN capabilities on existing hardware) and SMART (which optimises migration sequencing using reinforcement learning) are some of the frameworks that enable institutions to adapt modern features without replacing all their hardware at once ***(Tan et al., 2025).***

4.6 Comparative Summary

Criterion	Flat	Three-Tier	Spine-Leaf	Hybrid
Scalability	Poor	Good	Excellent	**Excellent**
East-West Performance	Poor	Moderate	Excellent	**Excellent**
Security / Segmentation	None	Good	Moderate	**Excellent**
Fault Isolation	None	Good	Good	**Excellent**
Legacy Compatibility	Yes	Yes	Partial	**Yes**
SDN / Automation	No	No	Partial	**Yes**

Table 3 - Summary Comparison of Network Models.

4.7 Recommended Model for CSDFL

The flat network model is rejected ***(Hirsi et al., 2024)***. It has a single broadcast domain; unrestricted lateral malware movement, failure to meet compliance requirements ***(Hirsi et al., 2024)***. According to Kurose & Ross (2022), Oppenheimer (2011) the Three-Tier Hierarchical model is the optimal modular architecture backbone for CSDFL; providing fault tolerance support as well as being in line with 10Gbps bandwidth demand and availability target of 99.9%. For resolving the bottlenecks processing intensive East-West traffic, Spine-Leaf architecture is incorporated into laboratory's micro-data centre providing a fixed two-hop path between any endpoints with lower and more predictable latency than hierarchical multi-hop designs ***(Al-Fares et al., 2008)*** within studies. SDN is overlaid to permit automatic evidence acquisition and real-time quarantine ***(Hirsi et al., 2024)***

The near-iron clad recommendation is then Hybrid Hierarchical: Fit for general lab connectivity, keep the excellent performing 10Gbps hierarchical backbone while moving forensic nodes into SDN-configured Spine-Leaf fabrics. It links legacy infrastructure outlays with modern-day high-performance functionality, all while still maximizing the dollar makeup in operation through right cost ***(Tan et al., 2025).***

5.0 Network Design Tools

It is noted that the first component of these design tools describes a software application which enables engineers and students to plan, simulate, and evaluate network topologies. In terms of the Network Development Life Cycle (NDLC), design tools provide an interface between architectural overview and deployment: they enable proposed designs to be tested, measured and adjusted without incurring the expense or risk of changing live infrastructure ***(Forouzan, 2013).***

Finding an appropriate tool to carry out the CSDFL LAN redesign means that it needs to be capable of modelling the proposed Hybrid Hierarchical architecture (i.e. VLANs, routing protocols, ACLs, QoS policies and redundancy need to be designed) while providing measurable performance characteristics in relation to the KPIs described in Section 3.4 in order to validate its design. Three popular tools – Cisco Packet Tracer; GNS3 (Graphical Network Simulator-3); and NetSim by Tetcos — are reviewed below.

5.2 Comparison of Three Tools

5.2.1 Cisco Packet Tracer

Cisco Packet Tracer (PT) was a complimentary and cross-platform visual networking simulation tool from Cisco Systems created for students enrolled into the Cisco Networking Academy***(cisco networking academy, 2024).*** It offers a drag-and-drop interface for building topologies, configuring simulated Cisco devices with CLI and monitoring protocol behaviour live. ***Mwansa et al. (2024)*** that it significantly enhances students' practical skills and theoretical understanding. ***Asadi et al. (2024)*** found that it also allows students to connect theory with practice without expensive physical hardware.

Packet Tracer, however, employs a limited set of command syntax for simulation purposes that does not cover the entire feature depth of any production IOS platform and is specific to Cisco devices only ***(Golightly et al., 2023).***

5.2.2 GNS3 (Graphical Network Simulator-3)

Moreover, GNS3 is a free and open-source network emulator that actually runs the Cisco IOS images on simulated hardware through Dynampis to deliver a more accurate environment than the simulation-based tools ***(GNS3 Documentation, 2024). Golightly et al. (2023)*** identifies it as well-known software for testing, emulating, configuring and troubleshooting virtual and physical networks at scale whose results closely match those

of real routers in terms of throughput, latency, and jitter measurements. GNS3 — the graphical network simulator that those familiar with more advanced Cisco routing and switching, for example CCNP or CCIE, will be well aware of GNS3 is actually multi vendor capable and can use QEMU, VMware and also Docker.

The main limitations are complexity and hardware demand. To legally license IOS images either physical Cisco hardware or a support contract is required, plus resource consumption increases non-linearly with topology size ***(GNS3 Documentation, 2024; Golightly et al., 2023).*** Mwansa et al. Tools that require substantial technical configuration are found to be less effective in structured educational environments ***(Kissinger et al.)***

5.2.3 NetSim by Tetcos

NetSim is a commercial entity -Discrete Event Network simulation laboratory of Tetcos created specifically for the delivery of higher education curriculum. Its Academic version provides Ethernet — TCP/IP — VLANs, ACLs, RIP, OSPF Wi-Fi and NAT through a no-scripting required drag-and-drop interface ***(Tetcos 2024a; Tetcos 2024b)*** One of the most distinguishing features is a built-in results dashboard that displays throughput, delay, jitter and packet delivery plots immediately after simulation run without any post-processing required ***(Tetcos 2024c).*** NetSim also has more than 35 pre-existing curriculum-aligned experiments — including VLAN configuration, link failure simulation and QoS — directly applicable to the CSDFL design ***(Tetcos, 2024a, 2024d).***

The downsides are lack of real IOS image emulation and more limited multi-vendor/SDN support relative to GNS3. As a commercial product, it needs a licensed academic installation to work and so is not available independently outside any university-based provision.

5.3 Justification of the Selected Tool

Table 3 below summarises the three tools against CSDFL's design requirements, based on the comparative analysis above:

Criterion	Cisco Packet Tracer	GNS3	NetSim (Tetcos)
Cost / Accessibility	Free (NetAcad account)	Free (IOS images required separately)	Licensed — available via university
Ease of Use	Very high — intuitive GUI	Low — significant setup complexity	High — drag-and-drop, no scripting

Criterion	Cisco Packet Tracer	GNS3	NetSim (Tetcos)
VLAN & ACL Support	Yes (simulated)	Yes (emulated, real IOS)	Yes (IEEE 802.1Q, ACL supported)
Routing Protocols	RIP, OSPF, EIGRP (simulated)	Full real IOS protocol suite	RIP, OSPF (simulated)
Performance Metrics	Basic — limited statistics	Moderate — requires external tools	Full built-in dashboard — auto-plots
Vendor Support	Cisco only	Multi-vendor	Generic/abstracted
Realism	Moderate — simulated commands	High — real IOS images	Moderate — discrete-event model
Hardware Requirements	Low	High	Low to moderate
Pre-built Experiments	Yes — activity-based	No — user-defined labs	Yes — 35+ curriculum-aligned
Suitability for Task 2 & 3	**Good**	**Moderate — too complex for module scope**	**Best fit for this module**

Table 4 - Selection Database Criteria for Network Tools

For this assignment, NetSim of Tetcos is the choosen tool for design and simulation. It is justified on three grounds.

First: Relevance and consistency with academic studies — This module, particularly its practical aspect, is based on a NetSim tool. As established in studies summarized by ***Tetcos (2024b),*** students who learn using a consistent toolset gain deeper procedural fluency and also have less cognitive overhead managing the interfaces of their tools — allowing more attention to focus on concepts for network design itself. This distinction would provide unnecessary inconsistency between Tasks 1, 2, and 3 given we are using a different tool to complete the design assignment.

Second, functional alignment: NetSim Academic version natively supports all protocols and related features needed to model CSDFL's Hybrid Hierarchical architecture – VLANs ***(IEEE 802.1Q)***, ACLs, OSPF, RIP, Ethernet switching and link failure simulation (Tetcos, 2024a, 2024d). These align exactly with the Technical Objectives listed in 3.3.

Thirdly, analyzing the performance: NetSim's collated results dashboard can yield throughput, delay, jitter, and packet delivery as statistics without any post-processing required ***(Tetcos, 2024c).*** This is crucial for both Tasks 2 and 3, where the delivered design concepts must be implemented and evaluated against the KPIs defined in Section 3.4 yet still — especially with regards to throughput targets of 1–10Gbps, latency below 5ms, and packet loss <1%.

GNS3 supports emulation of a real IOS, and GNS3 would be suitable for generating production-level design review models but comes at a high overhead in terms of compute resources and installation complexity which affects its suitability as part of an academic assignment with a low time requirement. Though highly accessible, Cisco Packet Tracer does not allow for the granular performance measurements necessary to analyze data in Task 3. NetSim is therefore the most suitable trade-off between capacity, accessibility and a product that aligns in with this module's assessment requirements.

6.0 Conclusion

Part 1 of the NDLC outlined in this document has laid the basis for the LAN redesign for Cyber Security & Digital Forensics Lab at St. John's University. Through four interrelated sections, it has set in place the analytical and strategic groundwork for a lab security-critical high performance research network solution.

The Top-Down design methodology was chosen as it allows the network architecture to be aligned with organisational visions from the beginning — this is especially pertinent due to the built-in security of CSDFL, scalable access for diverse groups and varied user-group access required. Ten objectives (five business and five technical) were defined based on academic literature as well as industry standards, along with measurable KPIs targeting throughput at 10Gbps, sub-5ms latency and an availability of 99.9%.

As a result of a comparative study of four network models — Flat, Three-Tier Hierarchical, Spine-Leaf and SDN — the Hybrid Hierarchical architecture was recommended. This solution retains the structured fault isolation and policy enforcement of the hierarchical model while integrating the low-latency, East-West performance of Spine-Leaf fabrics within forensic based micro-data centre, with SDN facilitating automated evidence management and real-time threat response.

Finally, three specific network design tools — Cisco Packet Tracer, GNS3 and NetSim — were evaluated in terms of the module's requirements. NetSim by Tetcos was chosen for its direct module integration, comprehensive remaining protocol and topology support as well as performance analytics dashboard being built into the tool for supporting directly design implementation and evaluation tasks that are part of Tasks 2 and 3 of this assignment. The Zero Trust microsegmentation controls discussed in Section 3.3.3 are a first implementation step; broader realization will follow an evolving maturity trajectory for multiple physical layers of the lab infrastructure following the CISA Zero Trust Maturity Model ***(CISA, 2023)*** and the Department of Defense Zero Trust Reference Architecture ***(DoD, 2022),*** as collegiate VLAN-based isolation policy is matured towards identity-driven continuous verification access enforcement at advanced dynamic routing capabilities within secure limit networks of resources in zones as architectures mature around this high-fidelity data analysis goal to be followed from foundational to efficient co-process control on computer architectures through into optimal cluster campaigns configured securely against shared workload abuses.

7.0 References

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